CFCMT: Structured Action-Cost Transfer for Traffic Signal Control

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***Abstract*—Road networks differ in junction layout, traffic demand, and available signal phases. A transferred controller must compare phase choices under these different conditions. We study a supervised transfer method that learns the cost difference between actions applied to the same simulated traffic state. A predictor restricted to local traffic features provides a reference, while a second predictor compares every pair of feasible phases. Development uses 18 networks in seven groups, followed by evaluation on two external networks. The final controller selects source-group components or a target anchor using target action labels and eight closed-loop development seeds, then freezes its predictors and execution guard. The complete comparison comprises 1,344 one-hour simulations across six controllers, four scenarios, and 56 paired seeds. Mean city-wise relative waiting decreases by 2.42% against MaxPressure (95% interval for the reduction, 0.66–4.58%), with 40 of 56 seed means favorable. Against PhasePressure, the reduction is 1.91%, although its interval crosses zero. Relative waiting also decreases by 2.30% against the legacy target anchor and 9.40% against the dense linear residual model. The selected controller uses an Atlanta component in Jinan and the target anchor in Los Angeles. These results support structured action prediction and the tested source-selection procedure on the two target networks, with all adaptation and selection completed before evaluation.**

***Index Terms*—Modelling and simulation, optimization and control, road transportation, traffic networks, traffic signal control, transfer learning.**

I. Introduction

A TRAFFIC signal controller that works at one junction may make poor decisions at another. The number and ordering of phases can change, downstream roads may have different capacities, and the same queue length can call for a different action under a different demand pattern. These changes make cross-network deployment a more demanding test than evaluating a controller on new random seeds of the network used for training. For an operator, the practical question is whether experience from other networks improves decisions at the new location after a limited amount of local preparation.

Many signal-control methods learn a policy or value function from interaction with a simulator. Others predict future traffic and choose an action from those predictions. In either case, the decision depends on how the available actions compare. Predicting the absolute traffic state accurately is useful only to the extent that the prediction distinguishes the consequences of competing phases. An error shared by every candidate can be large without changing the preferred action; a small error in the difference between two candidates can reverse the decision.

This observation motivates a narrower prediction task. We restore the same simulated traffic state, try each feasible phase, and measure the difference in the resulting short-horizon costs. The learner then predicts these within-state differences. This construction removes cost components common to all candidates and avoids assigning a fixed output coordinate to each phase identity. It does not remove errors that affect actions differently, and it does not establish that a short-horizon ranking will remain useful after the controller changes the traffic state through repeated decisions.

We investigate this approach through Causal-Factored Cross-City Model Transfer (CFCMT). Each source-group component combines two supervised predictors: a restricted predictor that compares each candidate with a pressure-based reference, and a pairwise predictor that compares every feasible pair. The second predictor is constrained so that reversing the order of a comparison reverses its sign. Target simulation data select how much weight to give each predictor. The fitted controller subsequently scores feasible phases from the observed traffic state, without online model updates or counterfactual simulation at each decision.

The study addresses three questions. First, can this representation improve phase selection across networks relative to a restricted predictor and an unrestricted residual baseline? Second, do improvements in offline action ranking translate into lower waiting time in closed-loop operation? Third, does the use of source-network examples improve on training from target data, once the target observations are held fixed? These questions require different comparisons. A model can outperform one residual estimator while remaining worse than a simple pressure rule, and a larger source-aware model can outperform a smaller target model without proving a benefit from source information itself.

We develop the method on 18 networks in seven groups and evaluate it on external Los Angeles and Jinan benchmark networks. The final controller selects source-group components or a target anchor using target development runs, then freezes both its predictor and execution guard. The main comparison tests this frozen source selector on 56 paired seeds. Earlier models and their separate validation experiments are reported in Appendix B.

The contributions are threefold. We formulate transfer around matched phase-cost differences and give the associated invariance and ranking properties. We implement a restricted predictor, an antisymmetric pairwise predictor, and a source-selection procedure that chooses a fitted component or target anchor before deployment. We then evaluate both their control performance and the extent to which gains can be attributed to source data, using pressure baselines and additional seeds on both targets. The results distinguish improvements in phase-

cost prediction from improvements in sustained traffic control.

II. Related Work

A. Traffic structure in signal control

Max-pressure control chooses service using upstream and downstream queue imbalance. Its stability results hold under the assumptions of the underlying traffic model [1]; they do not automatically extend to every microscopic implementation. PressLight and MPLight use pressure-related representations or objectives in learned controllers [2], [3]. The flipping- and rotation-invariant all-phase (FRAP) model represents pairwise competition between phases and exploits symmetry among traffic movements [4], while CoLight represents cooperation between intersections [5]. These studies motivate using traffic structure in the state and action representations. Our comparison also retains deterministic pressure policies, because improvement over another learned model alone does not establish an operational advantage.

B. Transfer across networks and simulators

MetaLight learns an initialization that can be adapted with target experience [6]. General Scenario-Agnostic (GESA) reinforcement learning provides a common representation for heterogeneous signal-control scenarios [7], while X-Light learns cross-city representations from decision trajectories [8]. CrossLight explicitly combines offline source training with online target fine-tuning [9]. Their target interaction and adaptation requirements differ from ours: we use labelled target simulator interventions and separate target development rollouts before deployment, then keep the deployed model fixed. Their published results cannot be substituted for a comparison under this information setting.

Simulator mismatch creates a related but distinct problem. Dynamics-Aware Hybrid Offline-and-Online Reinforcement Learning (H2O) and its extension, An Improved Framework for Hybrid Offline-and-Online Reinforcement Learning with Dynamics Gaps (H2O$^+$), combine fixed data with simulator interaction while addressing discrepancies in the dynamics [10], [11]. Grounded action transformation instead modifies actions to reduce the simulator-to-target discrepancy [12]. Our residual baseline is a supervised robust linear estimator. Our training and evaluation also use the same simulation engine, so the study concerns cross-network adaptation in simulation rather than transfer to a physical signal system.

C. Mechanisms and decision-relevant prediction

Work on causal representation learning motivates decomposing systems into mechanisms that can change separately across environments [13], [14]. Factored adaptation applies related ideas to non-stationary reinforcement learning [15]. Here, this motivation leads to predefined traffic features and controlled interventions within a simulator. We do not infer a causal graph from field observations. Our prediction target is the relative consequence of phase choices, and the final estimator uses gradient-boosted trees [16], [17]. This distinction matters: the evidence concerns the implemented predictors and control protocol, not the validity of a discovered causal structure.

III. Problem and Target Information

Consider a traffic network n with several signalized intersections. At a decision time, let s denote the observed traffic and signal state and let $\mathcal{A}(s)$ be the feasible green stages at the focal intersection. The controller chooses a stage every 10 s. A shared signal executor implements the requested stage while enforcing the signal program's minimum-green, yellow, and all-red requirements. This separates phase selection from the physical transition sequence.

For training, the complete Simulation of Urban MObility (SUMO) state is available [18]. Let g index a saved state s_g, with feasible actions $\mathcal{A}_g = \mathcal{A}(s_g)$, and let $Y_g(a)$ be the measured cost of applying action a at that state under a specified continuation policy. An *action group* contains the outcomes for all $a \in \mathcal{A}_g$. At deployment, these outcomes are unavailable: only state-derived features and the previously fitted model can be used to rank the actions. The objective is low waiting time over the full control run, assessed separately from the short-horizon training cost.

The available information consists of a source-network action bank, the target network and demand files, target action groups and development outcomes collected in simulation, and current target traffic observations during operation. The target simulator is not fitted to field trajectories. Nevertheless, its action outcomes are labelled target data. We call this *predeployment target adaptation*: all fitting and model selection finish before the evaluation rollouts. It neither requires online learning during those rollouts nor provides evidence of zero-shot or real-world transfer. Fig. 1 summarizes these information roles.

IV. Learning to Compare Phase Choices

The following score construction defines each source-group component. The final controller can also select a legacy target anchor, as described in Section IV-F.

A. Collecting matched action outcomes

After a 60-s warm-up, we save the simulator state and the states of all signal executors. Each candidate phase starts from an exact restoration of this snapshot. The focal signal requests that phase for one 10-s interval and then follows PhasePressure for five more intervals. Other signals follow PhasePressure throughout. Thus, each label measures one intervention followed by a common continuation, rather than a complete rollout of the learned controller.

The branch cost is the mean active-plus-pending vehicle load per controlled lane:

$$Y_g(a) = \frac{1}{60} \sum_{\tau=1}^{60} \frac{N_{\text{active}}(\tau) + N_{\text{pending}}(\tau)}{L_n}, \tag{1}$$

In (1), samples are one second apart and L_n counts distinct incoming lanes controlled by all signals in the network. $N_{\text{active}}(\tau)$ and $N_{\text{pending}}(\tau)$ count active and insertion-pending vehicles across that network at branch second τ, starting from s_g with action a. Pending vehicles are included so that restricting insertion does not appear to reduce the training cost.

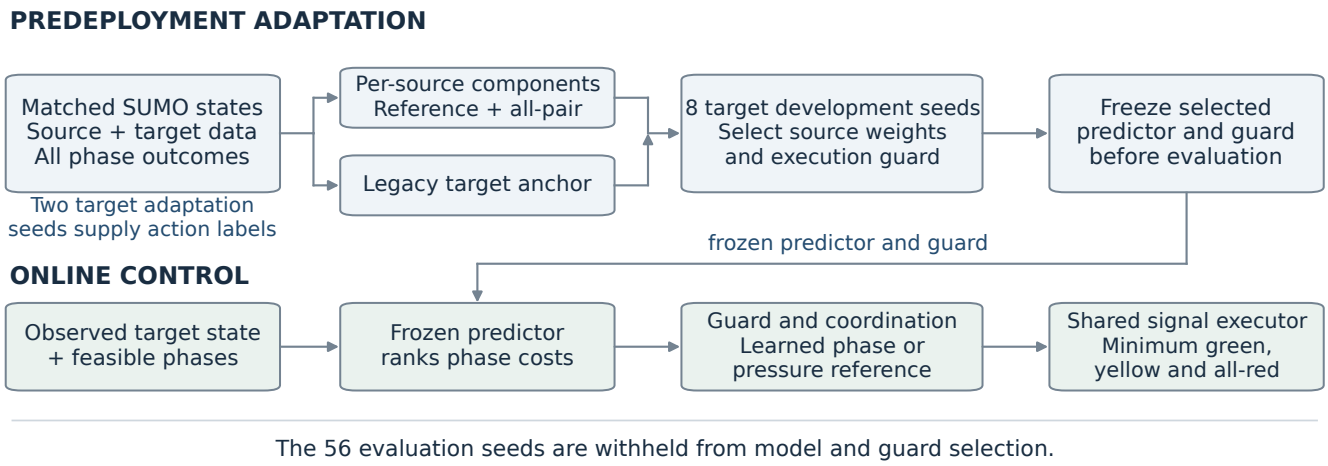


Fig. 1. Learning and deployment use different information. Matched simulator interventions provide source and target training outcomes. Two adaptation seeds supply target action labels; eight separate target development seeds select source weights and the execution guard. Deployment uses frozen predictors, the selected guard, sparse coordination, and current observations; evaluation outcomes are not used to select the controller.

Collection checks deterministic replay and rejects teleports. If any candidate has a junction collision, the whole action group is excluded; the retained groups therefore contain all candidate outcomes. This screening concerns the training groups and does not remove adverse runs from closed-loop evaluation.

A generalized pressure rule supplies a reference phase r_g. Its parameters are selected on source networks. The reference used for the external targets maximizes $q_{\mathrm{green}}(a) - 0.02o_{\mathrm{down}}(a)$, where the two features summarize the served-queue proxy and downstream occupancy. For lane ℓ, the queue proxy is $q_\ell = N_{\mathrm{halted},\ell} + 0.35N_{\mathrm{vehicles},\ell} + 0.025o_\ell$, where $N_{\mathrm{halted},\ell}$ and $N_{\mathrm{vehicles},\ell}$ count halted and total lane vehicles and o_ℓ is the simulator-reported lane occupancy; q_{green} sums the proxy over served incoming lanes. This reference defines the origin of the learned scores. It differs from the PhasePressure continuation, which maximizes the served-queue proxy alone.

Let $c_g(a) = Y_g(a) - Y_g(r_g)$. The normalized label and offline regret are

$$\sigma_g = \max\{\max_a Y_g(a) - \min_a Y_g(a), 10^{-8}\},$$
$$z_g(a) = c_g(a)/\sigma_g, \tag{2}$$
$$R_g(\widehat{a}) = \{Y_g(\widehat{a}) - \min_a Y_g(a)\}/\sigma_g. \tag{3}$$

The implemented scale also includes 10^{-6} times the 75th percentile of $|c_g(a)|$. Since the group includes $c_g(r_g) = 0$, this extra term cannot exceed a nonzero group range; (2) is the equivalent scale. No test outcome is needed at deployment: σ_g normalizes training labels and evaluation regret, not the model's runtime input.

B. Why compare costs within a state?

Suppose that $Y_g(a) = b_g + \mu_g(a) + e_g(a)$, where b_g is common to the candidates, $\mu_g(a)$ is the action-dependent cost component, and $e_g(a)$ is an action-dependent error. Subtracting the reference gives

$$c_g(a) = \mu_g(a) - \mu_g(r_g) + e_g(a) - e_g(r_g). \tag{4}$$

The common term cancels. Adding a constant to every outcome also leaves σ_g, the labels, and the optimal action unchanged. Positive rescaling preserves the action ordering; it preserves the normalized values provided neither the original nor the rescaled range activates the numerical floor. These elementary properties motivate the target representation. They do not imply that $e_g(a)$ is small or that phase effects are invariant between networks.

C. Restricted reference predictor

The first predictor, $A_g(a) = A_\theta(s_g, a)$, with fitted parameters θ, regresses $z_g(a)$ for non-reference actions and sets $A_g(r_g) = 0$. It uses 29 predefined features describing aggregate traffic state, queues served or stopped by the candidate and reference, downstream load, pressure, switching, clearance, and overlap with the current phase. Network identity and unrestricted static context are excluded from its prediction inputs. These restrictions encode an engineering hypothesis about useful traffic variables; they are not a learned causal graph.

We fit histogram-based gradient-boosted trees with squared-error loss. A training domain denotes one geographical or synthetic network group; target examples form an additional domain during adaptation. Base sample weights give each domain equal weight and each action group equal weight within its domain. We then multiply by $\mathrm{clip}(|z_g(a)|, 0.25, 4)$ and apply capped sign balancing so that the fit does not simply favor the more frequent sign of action improvement. Here clip limits its first argument to the stated interval. The resulting weights are normalized to mean one. Because of these adjustments, the final total weights need not remain exactly equal across domains.

D. Comparing every pair of phases

The second predictor uses a common state vector $x_g \in \mathbb{R}^{23}$ and a phase feature vector $\phi_g(a) \in \mathbb{R}^{40}$. The state features include local traffic, time, elapsed green, queue concentration,

and summaries of neighboring signals. Phase features describe demand served, downstream conditions, phase duration, transitions, and the affected road connections. For every unordered pair of actions a_i, a_j, training uses

$$\big([x_g, \phi_g(a_i) - \phi_g(a_j)],\ t_{gij} = z_g(a_i) - z_g(a_j)\big). \tag{5}$$

Square brackets in (5) denote vector concatenation. The reverse pair is added with the opposite target. Pair weights give equal total weight to domains, to groups within a domain, and to unordered pairs within a group. The same regression family as the reference predictor is used.

At prediction, we enforce antisymmetry explicitly:

$$\widehat{\delta}_{gij} = \frac{f_\psi(x_g, \phi_i - \phi_j) - f_\psi(x_g, \phi_j - \phi_i)}{2}. \tag{6}$$

Here f_ψ is the pairwise regressor with fitted parameters ψ, and ϕ_i abbreviates $\phi_g(a_i)$. With $K_g = |\mathcal{A}_g| \geq 2$, the action scores are

$$\begin{aligned} \widetilde{P}_g(a_i) &= \frac{1}{K_g - 1} \sum_{j \neq i} \widehat{\delta}_{gij}, \\ P_g(a_i) &= \widetilde{P}_g(a_i) - \widetilde{P}_g(r_g). \end{aligned} \tag{7}$$

An intersection with only one available action requires no comparison. If all pair predictions are exact, substitution in (7) gives

$$P_g(a_i) = \frac{K_g}{K_g - 1}\{z_g(a_i) - z_g(r_g)\}. \tag{8}$$

The factor is positive, so the action ranking is recovered. Its value is retained when combining the pairwise score with the reference predictor; the two score scales are not assumed identical. Appendix A gives the derivations of (4) and (8).

Both predictors use learning rate 0.05, 100 boosting iterations, at most 15 leaves per tree, and L_2 regularization 10, with early stopping disabled. Minimum leaf sizes are 12 for the reference model and 20 for the pairwise model. These settings are fixed before external evaluation.

E. Combining the predictors

Each anchored source component uses a convex combination,

$$S_g(a) = (1 - \alpha_g) A_g(a) + \alpha_g P_g(a), \qquad \widehat{a}_g = \operatorname*{arg\,min}_{a \in \mathcal{A}_g} S_g(a). \tag{9}$$

The final Jinan component uses constant $\alpha_g = 0.9$, inherited from the earlier city-level blend selection. These blend settings are not reselected for each source component or during the final evaluation.

The component predictors also return heuristic error scores. Let e_A and e_P be the 90th percentiles of absolute *in-sample* residuals for the two models, floored at 10^{-6}, and let d_g be the distance to the closest training-network context after standardization by the domain-centroid spread. For non-reference actions, define

$$\begin{aligned} u_{A,g}(a) &= e_A(1 + 0.25 d_g), \\ v_g(a_i) &= \frac{1}{K_g - 1} \sum_{j \neq i} \left| \widehat{\delta}_{gij} - (\widetilde{P}_g(a_i) - \widetilde{P}_g(a_j)) \right|, \\ u_{P,g}(a_i) &= (e_P + v_g(a_i))(1 + 0.25 d_g). \end{aligned} \tag{10}$$

Both error scores in (10) are set to zero at the reference action. Context is used in these diagnostics, not in direct score prediction. The quantity v_g measures disagreement between pair predictions and aggregated scores; because of the scaling in (8), it is not a pure measure of inconsistent preference cycles. Neither error score is an independently calibrated probability or confidence interval.

The component choice in (9) satisfies the general inequality $z_g(\widehat{a}_g) - \min_a z_g(a) \leq 2\|S_g - z_g\|_\infty$. It follows by adding and subtracting the predicted scores at the selected and optimal actions (Appendix A). This bounds component action regret in terms of prediction error, before source selection and execution guarding.

F. Source-group selection and deployment

The final source selector fits a separate instance of the anchored predictor for each source group, using that group's examples together with the target adaptation groups and inheriting the city-level blend. At zero source weight it uses the legacy target anchor. That model uses 117 features and retains its original domain-scaled labels: contrasts are divided by the median positive action-group range in the same training domain, with a numerical floor, then clipped to $[-5, 5]$. Its target regressor uses learning rate 0.05, 80 boosting iterations, at most seven leaves, minimum leaf size four, and L_2 regularization 16. Its target prediction weight is numerically almost one; a retained source prior can still resolve ties.

Selection uses eight target development seeds in three stages. First, each of seven source groups is tested at source weight 0.75 against the target anchor. The selected source is then tested at weights $\{0, 0.25, 0.5, 0.75, 1\}$. A source candidate is eligible only with at least 0.5% mean waiting improvement, at most 1% worst-seed regression, and no added collision or teleport incidents per seed. Eligible candidates minimize mean relative waiting change plus half its standard deviation and the positive part of the worst relative change; otherwise selection returns the target anchor. Finally, guard profiles with risk multipliers 0, 0.25, and 0.5 are compared with the selected unguarded controller. Eligible guards do not increase aggregate incidents or exceed 1% worst-seed regression, and must reduce incidents or improve mean waiting by at least 0.5%. Selection prioritizes fewer incidents, then the same waiting score; the unguarded controller remains available. All these choices precede the final evaluation.

The resulting Jinan controller uses the Atlanta component at source weight one and guard multiplier 0.5. Its error score is $(1 - \alpha_g)u_{A,g}(a) + \alpha_g u_{P,g}(a)$. A learned phase replaces the generalized-pressure reference only when its predicted contrast plus 0.5 times this error score is negative and its normalized pressure deficit is at most one. The deficit is $\max\{\Pi(r_g) - \Pi(a), 0\} / \max\{|\Pi(r_g)|, 1\}$, where Π is the generalized-pressure score defined above. Los Angeles selects zero source weight and the legacy target anchor without this guard.

Learned departures from the pressure reference use sparse coordination. At each decision, a greedy selection admits mutually nonconflicting proposals on the signal conflict and

TABLE I
NETWORK INVENTORY

Group	Networks	Signals	Lanes
Atlanta	1	5	55
Cologne	3	12	252
Hangzhou	6	36	532
Ingolstadt	3	32	2,227
New York	1	196	2,562
Synthetic	2	32	360
Salt Lake City	2	4	88
Source total	18	317	6,076
Los Angeles (target)	1	4	157
Jinan (target)	1	12	186

routing graph. A five-interval cooldown gives a minimum 60-s gap between interventions at the same or conflicting signals, with 10-s decisions. Proposals that are blocked or not selected use the pressure reference. Pure MaxPressure and PhasePressure use the common phase executor without this learned-intervention coordinator. The predictors, source weights, guard, and coordination settings remain fixed throughout evaluation. Appendix B reports the earlier pooled-source controller separately.

V. EVALUATION DESIGN

A. Networks and simulator

Experiments use SUMO 1.22.0 through `libsumo` [18]. The source collection contains 18 networks, 317 controlled signals, and 6,076 non-internal lanes, grouped as in Table I. There are six geographical groups and one synthetic group. Holding out a geographical group removes all of its subnetworks from source fitting. The collection draws on public signal-control benchmarks, including Reinforced Signal Control (RESCO) and LibSignal [19], [20].

The external targets are a four-signal Los Angeles corridor from LibSignal and a 12-signal Jinan network from the CoLight benchmark [5]. Jinan has three demand files on the same geometry. These inputs were distributed in CityFlow format [21] and converted to SUMO, preserving demand, route continuity, phase identities, and protected versus permissive movements. A pre-evaluation execution check was used to detect conversion defects. These are benchmark representations of parts of cities, not calibrated city-wide digital twins.

B. Separating development, adaptation, and tests

Development uses source seeds 2027 and 3037 for fitting and seed 4047 for held-out evaluation. The leave-one-group-out exercises allow up to 100 adaptation groups from the held-out network group; they are tests of adaptation rather than pure zero-shot transfer. For the final external models, the source bank includes all three source seeds. Thus, 4047 is held out for development assessment but is not withheld from the later source-bank refit.

Target adaptation seeds 5057 and 6067 yield 165 complete action groups in Los Angeles and 911 in Jinan. Each group contains multiple alternative-action outcomes and should not be interpreted as one passive observation. For the earlier development tests in Appendix B, seed 8171 supplies 79 Los Angeles and 457 Jinan offline test groups, and the initial closed-loop comparison uses seeds 8081 and 9091. An earlier diagnostic seed, 7079, influenced method development and is excluded from the reported tests. Final evaluation outcomes are not used to tune the predictors, source weights, or guard.

Each closed-loop run lasts 3,600 s in total, including a 60-s warm-up. The final comparison uses four target scenarios and 56 paired seeds. The two cities receive equal weight; Jinan's three demands are first averaged within that city. The original seven-controller, two-seed comparison and its separate confirmation tests are reported in Appendix B.

For the final source selector in Section IV-F, static network context is repaired by tracing controlled movements through intermediate links to the nearest downstream signals, and context-dependent analytic priors are recomputed. The recorded action features and outcome labels remain unchanged. Models use all 165 Los Angeles or 911 Jinan adaptation groups. Eight separate target development seeds select the source group, source weight, and guard. Selection returns the legacy target anchor without a guard for Los Angeles and the Atlanta component at weight one for Jinan, with constant blend weight 0.9 and guard multiplier 0.5. The target-anchor and equal-source comparators receive the same city-specific guard and sparse coordination as the selected controller. The latter averages predictions of the seven source-group components with equal weights. All choices are frozen before 56 paired evaluation seeds. The original five-controller matrix gives 1,120 one-hour runs. MaxPressure is subsequently added as a fixed baseline on the same 56 seeds and four scenarios, giving 224 further runs and 1,344 in total. It uses the existing MaxPressure implementation, with no model refitting or parameter selection. All six methods share network files, phase executor, and decision timing. This protocol uses target closed-loop development outcomes in addition to the two adaptation seeds.

C. Comparators and common execution

The final comparison includes the frozen CFCMT selector, the legacy target anchor, an equal-weight mixture of the seven source components, a frozen dense linear residual model, PhasePressure, and MaxPressure. The implemented MaxPressure subtracts the downstream-queue proxy from the served-queue proxy. PhasePressure uses the served-queue proxy alone. The development comparisons additionally include fixed-time control, an uncorrected analytic scoring rule, and the restricted reference predictor alone (Appendix B).

The dense linear model fits a correction to the analytic prior using the full local-feature and context vector. It uses domain-balanced robust ridge regression, with L_2 penalty 3, Huber parameter 1.5, and three robust-fitting iterations. It originally receives the same matched action groups and target refit data as the pooled-source controller and remains frozen for the final comparison. The source-selection components instead use individual source groups and the same target adaptation data. The methods also differ in function class,

normalization, and weighting. The comparison therefore tests complete estimator configurations; it does not isolate the effect of feature restrictions at matched model capacity.

All controllers share network and demand files. Learned and pressure controllers use the same candidate phase identities and phase executor; fixed-time control retains its native SUMO program. The executor reads the original program's positive minimum-green duration; otherwise it uses up to 5 s, bounded by the phase duration and at least 1 s. Yellow and all-red durations use the corresponding program medians, with defaults of 3 s and 1 s. All-red is extended in one-second increments while the junction interior is occupied. These constraints are implementation safeguards, not a proof of collision-free control.

Published reinforcement learning methods were also run under their original RESCO or LibSignal interfaces. Their observations, rewards, timing, and training procedures differ from this comparison. We therefore do not insert their scores into the matched six-controller table or claim superiority over those methods.

D. Outcomes and uncertainty

The primary outcome is average waiting time over all vehicles that depart during a run, including waiting accrued by unfinished trips up to the fixed endpoint. Waiting records and cumulative departure and arrival counts include the warm-up period. The measure avoids reporting only completed trips, but remains finite-horizon and excludes vehicles that never depart. System vehicle-hours sum one-second active-plus-pending vehicle counts from the warm-up endpoint through the simulation endpoint and divide by 3,600; the 60–3,600 s endpoints give 3,541 samples. Demand completion is $C_{\mathrm{arr}}/(C_{\mathrm{dep}} + N_{\mathrm{pending}}(T))$, where C_{arr} and C_{dep} are cumulative arrivals and departures at the simulation endpoint $T = 3{,}600$ s. Completed-trip waiting and simulator incidents are also reported. Together, these measures expose changes in insertion, network loading, and unfinished traffic.

Offline regret follows (3); its development intervals use 10,000 paired group-bootstrap replicates. Additional statistical procedures for the development comparisons are given in Appendix B.

For the 56-seed source-selection comparison, waiting is first averaged across scenarios within each city and seed. Relative change is then calculated against the comparator within each city, followed by equal averaging across cities and seeds. This registered mean of city-wise ratios differs from both the ratio of pooled waiting means and (11). The 95% intervals use 20,000 paired seed-block bootstrap replicates. The original four-comparator audit uses Holm correction over 12 two-sided Wilcoxon tests. With MaxPressure added, the expanded report applies Holm correction over all 15 city or equal-city comparisons against five comparators. The original intervals are retained; the three supplemental MaxPressure intervals use NumPy's permuted congruential generator (PCG64) with seeds 20261030, 20261031, and 20261032 for Los Angeles, Jinan, and the equal-city effect, respectively, fixed before the new runs. All 56 seeds are retained.

VI. Results

A. Frozen source selection against pressure control

The final comparison comprises 1,344 completed runs across six methods, four scenarios, and 56 paired seeds. The frozen CFCMT selector achieves the lowest equal-city mean waiting in Table II. Its registered mean city-wise relative change against MaxPressure is -2.42%, with a paired 95% interval of $[-4.58, -0.66]\%$. Waiting improves in 40 of 56 equal-city seed means, and the Holm-adjusted Wilcoxon p-value is 0.0058. Jinan improves by 2.10%, with all 56 seed means favorable and interval $[-2.35, -1.85]\%$ for relative change. Los Angeles improves by 2.74% on average, with 33 favorable seeds and interval $[-7.05, 0.77]\%$.

Against PhasePressure, the registered mean change is -1.91%, with interval $[-4.42, 0.08]\%$ and 34 favorable seed means. The city-specific changes are -3.73% for Los Angeles and -0.10% for Jinan. Its equal-city interval crosses zero. Figure 2 retains all comparator intervals and all 56 seed effects against both pressure rules, including the large favorable tails.

B. Source selection and learned comparators

Against the legacy target anchor, equal source weights, and the frozen dense linear baseline, the corresponding relative changes are -2.30%, -2.46%, and -9.40%, respectively; all three paired intervals exclude zero. The selected controller exactly matches the target anchor in Los Angeles and improves on it by 4.60% in Jinan, with all 56 Jinan seed means favorable. The raw equal-city waiting means in Table II and the registered mean relative changes summarize the same complete runs with different weighting: taking a ratio after pooling the waiting values gives greater weight to large Los Angeles values.

Appendix B reports the earlier models' full offline and closed-loop comparisons, the ten-seed frozen-model confirmation, and the separately selected Jinan controller. Those experiments use different fitted models or selection procedures and are not pooled with the final source selector.

VII. Discussion

A. What the results establish

The frozen source selector improves the paired-seed waiting outcome over MaxPressure in the two-network comparison. Its Jinan improvement over the legacy target anchor and exact fallback to that anchor in Los Angeles support the tested source-selection procedure. The favorable PhasePressure point estimate gives a less conclusive comparison because its interval crosses zero.

The main comparison and the development tests in Appendix B support the tested predictor configurations over the dense linear residual baseline. They do not show that every component is necessary. In particular, the gain may reflect the nonlinear regressor, action normalization, feature restrictions, pairwise learning, weighting, or their interaction. A capacity-matched nonlinear baseline and controlled component ablations are needed before attributing the effect to a specific architectural choice. The reference-only comparison provides

TABLE II
FROZEN SOURCE SELECTION: FOUR SCENARIOS AND 56 PAIRED SEEDS

	Waiting, all departed vehicles (s)			System	Demand
Controller	Los Angeles	Jinan	Equal-city mean	vehicle-hours	completion
CFCMT, frozen selector	201.68	89.64	**145.66**	585.42	0.895
Legacy target anchor	201.68	93.97	147.83	588.48	0.894
Equal source weights	212.45	89.62	151.03	586.27	0.895
Frozen dense linear residual	211.73	104.59	158.16	613.28	0.876
PhasePressure	231.36	89.73	160.55	600.47	0.878
MaxPressure	221.49	91.57	156.53	604.78	0.879

Each row contains 224 complete runs. These are raw equal-city means; registered relative effects first form ratios within each city and seed, then average cities and seeds. MaxPressure is a later fixed-baseline supplement on the same cohort, with all models and execution settings unchanged. The target anchor retains a source prior and is not strictly source-free.

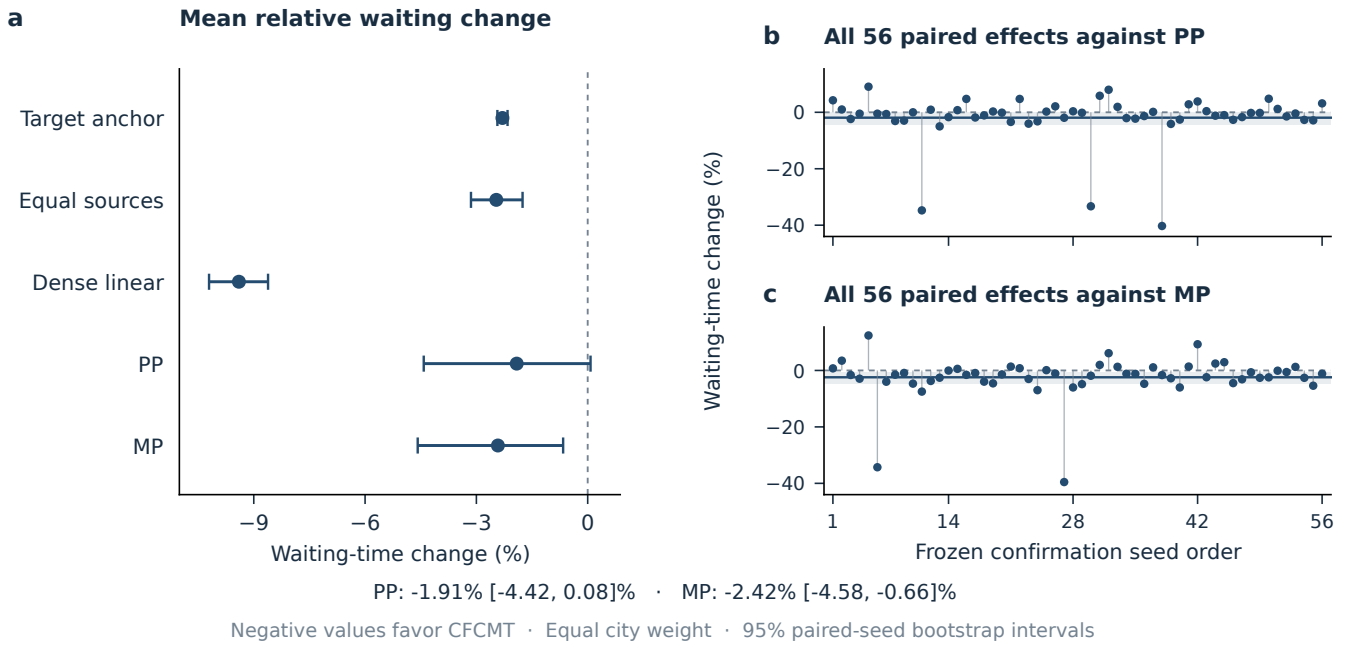


Fig. 2. Frozen source selection on 56 paired seeds. (a) Mean city-wise relative waiting changes and 95% paired bootstrap intervals against all five comparators. (b, c) All 56 equal-city relative effects against PhasePressure (PP) and MaxPressure (MP), in the frozen seed order. Solid lines mark means, shaded bands their intervals, and dashed lines zero. Negative values favor the selected controller. Scenarios are averaged within each city before forming relative changes; cities receive equal weight.

one such partial ablation, but its closed-loop interval remains inconclusive.

B. Why offline ranking may not predict control performance

There are two changes between the supervised task and deployment. First, the labels measure one focal phase choice followed by PhasePressure, with other intersections also using PhasePressure. The deployed controller repeatedly chooses learned phases at multiple intersections. It can therefore visit states and joint action combinations that differ from those under the label continuation. Second, the training target is a 60-s vehicle-load measure, whereas evaluation measures waiting and completion over an hour. An action can improve the former without improving the latter. These differences offer plausible explanations for the imperfect relationship between offline regret and closed-loop waiting, but the present experiments do not identify their individual causal contributions.

C. Scope of the evidence

This result includes eight target closed-loop development seeds and changes both model fitting and execution. The anchor and source component also differ in model capacity. It therefore measures the complete selection procedure, rather than an isolated benefit of source data or the original controller alone. The city-specific MaxPressure interval still crosses zero for Los Angeles; the confirmed aggregate effect is not a separate confirmation in every city.

The external study contains two geometries. Three demand files for Jinan and additional random seeds increase coverage within those geometries; they do not increase the number of independent cities. Target-network diagnostics were used during development, although the reported final seeds were withheld. The simulator is not calibrated to field trajectories, and both adaptation and evaluation use the same physics engine. We therefore cannot infer physical deployment per-

formance, robustness to a different simulator, or effectiveness under detector failure and unobserved demand regimes.

The word "causal" in the method's name refers to controlled simulator interventions and predefined input restrictions. It does not claim observational causal discovery or invariance in the physical network. Similarly, heuristic error scores do not certify safety. Incident counts in the development comparisons (Appendix B) demonstrate why the original zero-incident runs must not be promoted into a general safety statement.

Finally, same-protocol comparisons with transfer methods such as CrossLight, X-Light, GESA, and MetaLight remain missing. An informative next evaluation would match model capacity and target information, use the same signal executor, and include additional networks withheld from method development. It should test target-only and source-aware training with identical architectures and measure both short-horizon action quality and sustained closed-loop behavior. That design would distinguish a useful representation from a demonstrable benefit of transferred data.

VIII. Conclusion

We presented a supervised method for adapting signal control across road networks by learning differences between feasible phase choices. A restricted reference predictor and an antisymmetric pairwise predictor are combined using target simulation data, then deployed without online refitting. The final controller selects source components or a target anchor using eight target development seeds before freezing its predictions and execution guard. On 56 paired seeds across two target networks, it reduces mean city-wise relative waiting by 2.42% against MaxPressure, with a 95% reduction interval of 0.66–4.58% and 40 favorable seed means. It also lowers waiting relative to the tested target anchor and dense linear residual baseline. The PhasePressure point estimate is favorable, but its interval crosses zero. The evidence supports the complete source-selection procedure on these fixed networks and motivates capacity-matched tests of the source data's contribution on additional independent networks.

Data and Code Availability

The implementation and exported analysis tables for the initial experiments are available in the CFCMT repository, at commit `61e8a81`, including the `paper_tsc/source_data` directory.[1] That commit also contains source-component fitting and deployment code. The accompanying source-data files additionally contain the 80 ten-seed confirmation runs and the 1,344 frozen source-selection and baseline runs, with paired summaries; these results are not part of that earlier commit. Supplemental MaxPressure execution and analysis scripts are in `experiments/tsc_v93_max_pressure_20261030`. The exports support the numerical summaries reported here. Full simulation logs, trained models, and some aggregated evaluation records remain in the execution archive and are not all included in this repository snapshot. The release therefore does not yet provide an end-to-end reproduction package. The benchmark inputs retain their upstream licences.

[1] https://github.com/erzhu419/CFCMT, commit `61e8a81`.

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Appendix A
Proofs of the Score Properties

Common-offset cancellation. Subtracting $Y_g(r_g)$ from $Y_g(a)$ cancels b_g and gives (4). Adding a common constant leaves every contrast and the group range unchanged. Multiplying all outcomes by a positive constant multiplies both by that constant, so normalized labels are unchanged when the numerical floor is inactive. Action ordering is preserved in either case.

Pairwise ranking recovery. Swapping i and j in (6) negates the prediction. For exact pair predictions,

$$\sum_{j\neq i}\{z_g(a_i)-z_g(a_j)\} = K_g z_g(a_i) - \sum_{j=1}^{K_g} z_g(a_j).$$

Dividing by K_g-1 and subtracting the reference score gives (8). The common subtraction and positive factor preserve the minimizers, and $P_g(r_g)=0$.

Ranking error. For the ranking bound, let $a_g^\star$ minimize z_g. By the definition of $\widehat{a}_g$ in (9),

$$\begin{aligned} z_g(\widehat{a}_g) - z_g(a_g^\star) &= z_g(\widehat{a}_g) - S_g(\widehat{a}_g) \\ &\quad + S_g(\widehat{a}_g) - S_g(a_g^\star) \\ &\quad + S_g(a_g^\star) - z_g(a_g^\star) \\ &\le 2\|S_g - z_g\|_\infty. \end{aligned}$$

Here $\|S_g - z_g\|_\infty$ is the largest absolute score error over the feasible actions. This is an algebraic bound; it assumes no error guarantee for an unseen network.

Appendix B
Development Models and Separate Comparisons

These experiments concern the original pooled-source controller and a separately selected Jinan model. Each comparison retains its own fitted controller, target data, and evaluation protocol. The final frozen source-selector comparison is reported in Section VI.

A. Offline phase selection

In the seven development groups, mean normalized regret decreases from 0.24142 for the restricted reference predictor to 0.21767 for the selected blend, a reduction of 9.84%. Six groups improve. Hangzhou worsens by 0.00388, showing that the average improvement does not hold uniformly across groups. These results concern the development adaptation exercises, not seven independent external cities.

TABLE III
ORIGINAL POOLED-SOURCE CONTROLLER: EXTERNAL OFFLINE REGRET

Target	Reference	Original CFCMT	Reduction
Los Angeles	0.14824	0.14501	2.18%
Jinan	0.43615	0.39208	10.10%
Equal-city mean	0.29219	0.26854	8.09%

Lower regret is better. Paired 95% intervals for absolute regret reduction are $[-0.06747, 0.07289]$ for Los Angeles and $[0.00469, 0.08293]$ for Jinan. Test groups come from one held-out seed.

The external offline test is summarized in Table III. Equal-city regret decreases from 0.29219 to 0.26854, an 8.09% reduction. Los Angeles improves by 2.18%, but its paired improvement interval crosses zero. Jinan improves by 10.10%, with a positive group-bootstrap improvement interval. Correct selection of the minimum-cost action increases from 78.5% to 81.0% in Los Angeles and from 29.6% to 34.2% in Jinan. The unequal accuracies also indicate that the two targets differ substantially in ranking difficulty.

B. Initial comparison of the original controller

Table IV reports all seven controllers. The original controller achieves 149.17 s mean waiting, compared with 159.15 s for the dense linear residual model. The reduction is 9.98 s, or 6.27%, with a paired bootstrap interval of $[2.01, 17.19]$ s. Waiting decreases in both targets: by 8.46 s in Los Angeles and 11.50 s in Jinan. Completed-trip waiting and system vehicle-hours also decrease in this comparison.

Relative to the simulator prior and reference predictor, the proposed controller reduces the waiting point estimate by 4.81% and 2.45%, respectively. Their intervals cross zero. Thus, the initial closed-loop evidence is clearest for the comparison with the dense linear residual. Since both methods use matched action differences, that gain cannot be credited to reference subtraction alone. The different regressors and weighting procedures also prevent an isolated claim about causal feature restrictions.

The two pressure controllers achieve the lowest equal-city waiting times in this initial comparison. MaxPressure outperforms the original CFCMT by 7.43 s and PhasePressure by 5.55 s. Both also reduce system vehicle-hours and improve completion. Their superiority is visible in each target, so it is not an artifact of averaging a Los Angeles gain against a Jinan loss.

All 56 runs complete without teleports. The eight proposed-controller runs contain no consolidated junction-collision incidents. This small initial sample is insufficient to establish a safety advantage, especially given the larger comparison below.

C. Independent confirmation with frozen models

We evaluate the same proposed and dense linear residual models on ten new paired seeds across the original four scenarios. All 80 runs complete without refitting or parameter tuning. Equal-city waiting decreases from 158.67 to 150.84 s, a reduction of 7.83 s or 4.93% (Table V). The paired 95% interval is 4.11–11.85 s for the absolute reduction and 2.64–7.31% for the relative reduction. The prespecified confirmation criterion is met, and nine of ten equal-city seed means improve. Los Angeles improves by 5.43 s on average, with six favorable seeds; Jinan improves by 10.22 s, with all ten favorable (Fig. 3). Completed-trip waiting decreases from 147.49 to 132.88 s, system vehicle-hours from 616.49 to 596.15, and completion increases from 87.59% to 88.66%. All runs have zero teleports. These results confirm the frozen controller's advantage over the tested linear residual baseline on new seeds of the same networks.

D. Fresh seeds reveal different transfer outcomes

The original seven-controller comparison did not include the existing target-dominated model as a closed-loop baseline. Adding that model afterwards to the original eight scenario–seed cells gives 149.04 s mean waiting, almost identical to CFCMT's 149.17 s. Because the addition followed inspection of the main results, this comparison is descriptive.

We subsequently compare the unchanged controllers on 64 fresh seeds, one Los Angeles scenario and three Jinan scenarios, for 512 runs. The comparison model's deployed prediction weight is numerically entirely on target data, but a retained source prior can resolve exact score ties. We therefore call it the *target-dominated comparator*, not a strictly source-free model. Both controllers use the same target adaptation groups and execution rules.

The equal-city mean waiting is 155.14 s for CFCMT and 149.09 s for the comparator. The prespecified mean seed-level relative change in (11) is $+4.25\%$, with interval $[-0.77, 13.21]\%$. Only 29 of 64 seed means improve. The city effects have opposite signs: Jinan improves by 3.04%, with all 64 seed means favorable, while Los Angeles worsens by 7.87%, with 22 favorable seeds.

The largest Los Angeles failure increases waiting from 194.74 to 975.27 s and reduces completion from 0.919 to 0.270. It has neither a teleport nor a collision and remains in the analysis. Across the full comparison, collision incidents total 25 for CFCMT and 15 for the comparator. The prespecified requirement for joint benefit—including nonpositive average and interval upper bound, no city regression, at most 5% worst-seed regression, and no excess incidents—is not met. More repetitions therefore do not confirm a general benefit from the original source-aware controller.

E. A separately selected Jinan controller

Following that result, a subsequent experiment refits the models and selects among seven source groups using only target action labels. It corrects static network context by tracing controlled movements through intermediate road links to the nearest downstream signals, then recomputes context-dependent analytic priors. Recorded action features and outcome labels are unchanged. Five folds, blocked by scenario and seed, evaluate a balanced set of 100 target action groups and source weights $\{0, 0.25, 0.5, 0.75, 1\}$. A source is used only if mean regret improves by more than 0.5%, its one-sided 95% upper bound for change is below zero, and no fold

TABLE IV
ORIGINAL CONTROLLER: INITIAL OUTCOMES OVER FOUR SCENARIOS AND TWO SEEDS

	Waiting, all departed vehicles (s)			Completed-trip	System	Demand
Controller	Los Angeles	Jinan	Equal-city mean	waiting (s)	vehicle-hours	completion
Fixed time	507.33	169.84	338.58	308.72	885.18	0.661
Dense linear residual	212.88	105.41	159.15	147.47	618.15	0.870
Simulator prior	219.17	94.26	156.71	138.68	619.66	0.877
Reference predictor	206.78	99.05	152.91	138.40	606.30	0.882
CFCMT, original	204.42	93.91	149.17	133.06	594.14	0.889
PhasePressure	197.51	**89.73**	143.62	**125.32**	583.05	**0.898**
MaxPressure	**191.89**	91.58	**141.74**	135.42	**580.29**	0.896

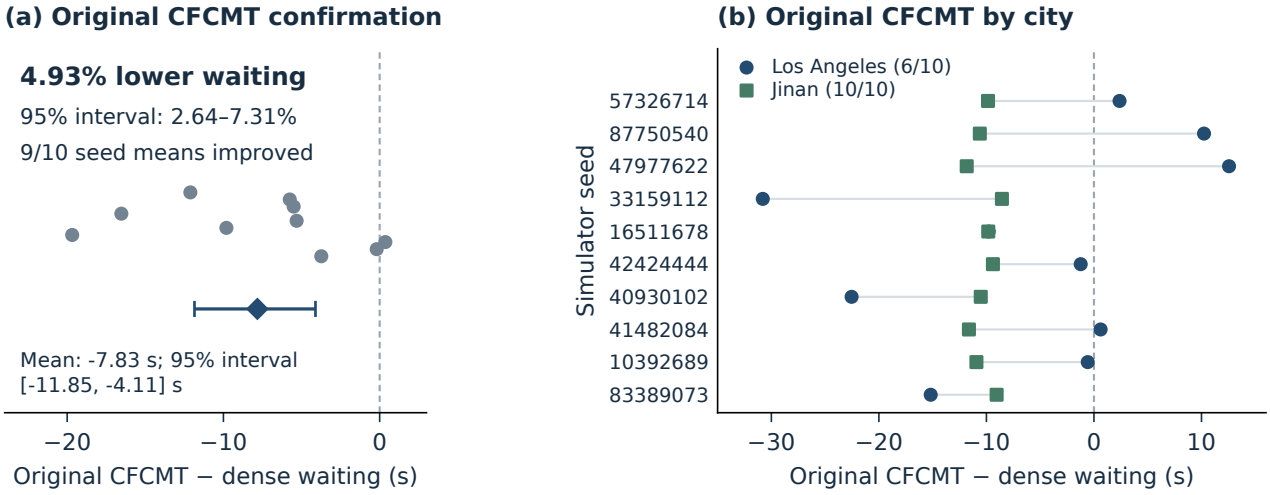


Fig. 3. Original pooled-source controller: frozen-model confirmation on ten new paired seeds. (a) Grey points show equal-city waiting differences for all ten seeds, with vertical offsets for visibility; the diamond shows the overall mean with its registered 95% interval from 20,000 paired seed-block bootstrap replicates. (b) City-specific differences retain the same seed pairing. Negative values favor the original CFCMT. The interval describes seed variation within the two fixed target networks.

TABLE V
ORIGINAL POOLED-SOURCE CONTROLLER: TEN-SEED CONFIRMATION

	Waiting (s)			
Target	Dense	Original	Reduction	Improved
Los Angeles	213.07	207.64	2.55%	6/10
Jinan	104.27	94.05	9.80%	10/10
Equal-city mean	158.67	150.84	4.93%	9/10

Waiting includes unfinished trips. Jinan demands and then cities are weighted equally. The overall relative reduction is computed from the two macro means, with a paired 95% interval of 2.64–7.31%. Improved counts refer to seed-level waiting means.

worsens by 1% or more. Otherwise the procedure returns the target-only model. Before its new evaluation runs, it selects no source for Los Angeles and the Hangzhou component at weight one for Jinan.

On 56 further Jinan seeds and three demand scenarios, the selected controller reduces waiting by 4.53%, with paired interval $[4.28, 4.79]\%$; all 56 seed means improve. The experiment comprises 336 runs. This is a repeatable improvement over a lower-capacity model fitted only on target examples. Both arms use the same pressure-reference guard: a proposed phase can replace the reference only when its predicted cost contrast plus 0.5 times its error score is negative and its normalized pressure deficit is at most one. Otherwise, the reference phase is used. The deficit is $\max\{\Pi(r_g) - \Pi(a), 0\}/\max\{|\Pi(r_g)|, 1\}$, where Π is the generalized-pressure score and r_g is its reference phase in the current state. The multiplier 0.5 is inherited from earlier development. This comparison cannot separate the value of source data from the model change and cannot be pooled with the original controller's results. Table VI summarizes these two protocols.

F. Additional development diagnostics

The pressure comparisons keep this limitation visible. A separate later experiment using a 120-s predictor and a rate-limited executor also fails to establish an advantage over PhasePressure on 64 fresh Jinan seeds: its relative waiting change is -0.043%, with interval $[-0.449, 0.370]\%$. That controller differs from CFCMT and is not an additional row in Table IV. Further source-selection experiments on the seven development groups either retain harmful group-level effects or return target-only predictions. They provide no basis for

TABLE VI
FRESH-SEED COMPARISONS UNDER SEPARATE PROTOCOLS

Controller and target	Seeds	Change	Improved
Original, two-city mean	64	$+4.25\%$	29/64
Original, Los Angeles	64	$+7.87\%$	22/64
Original, Jinan	64	-3.04%	64/64
Revised controller, Jinan	56	-4.53%	56/56

Negative change means lower waiting. The original controller is compared with the target-dominated model; the revised controller uses a smaller, strictly target-only comparator. The primary 95% intervals are $[-0.77, 13.21]\%$ for the two-city mean and $[-4.79, -4.28]\%$ for the revised controller. The rows are not estimates for one unchanged method.

replacing the negative cross-network result with a general transfer claim.

G. Uncertainty for development comparisons

Offline regret follows (3). Group-level uncertainty uses 10,000 paired bootstrap replicates. For the initial closed-loop comparison, scenarios and seeds are paired between controllers. The 10,000-replicate hierarchical bootstrap resamples cities, then scenarios, then seeds. With only two target networks and two seeds per scenario, these intervals describe the observed experiment; they do not establish performance over a population of cities.

For the independent dense-baseline confirmation, ten new seeds are fixed before execution, excluding previously used and reserved seeds. Both models, network inputs, signal executor, and timing remain unchanged, giving 80 runs on the original four scenarios. No refitting, new target labels, or parameter tuning is used. Within each seed, the three Jinan demands are averaged equally, followed by equal city weights and then equal weights across seeds. The primary effect is the difference between the two overall waiting means; relative change divides this difference by the dense mean. It is not the mean of seed-level percentages. The 95% percentile interval uses 20,000 paired bootstrap replicates with NumPy PCG64, initialized with 20261022. Each replicate resamples ten complete seed blocks, keeping both controllers and all four scenarios together. Cities are not resampled. Confirmation requires all 80 valid records, a negative mean waiting difference, and a negative upper interval endpoint. All ten seeds are retained, and these runs are not pooled with the other experiments.

In the 64-seed target-dominated comparison, each seed $k = 1, \ldots, N$, with $N = 64$, produces an equal-scenario mean for each city and then an equal-city mean $\overline{W}_k^{\pi}$ for controller π. Relative change is

$$\rho_k = \frac{\overline{W}_k^{\text{proposed}} - \overline{W}_k^{\text{comparison}}}{\overline{W}_k^{\text{comparison}}}, \qquad \overline{\rho} = \frac{1}{N}\sum_{k=1}^{N} \rho_k. \quad (11)$$

Negative values indicate lower waiting. The mean of these ratios is distinct from the ratio of the two pooled waiting means. Fresh-seed intervals resample complete paired seeds. No adverse evaluation seed is removed.

H. Original local blending rule

Candidate combinations include constant choices $\alpha_g \in \{0, 0.1, \ldots, 1\}$ shared across groups, and a local rule that limits departures from A_g.

Let a_A and a_P minimize A_g and P_g, respectively, and write $\text{range}(v) = \max_a v(a) - \min_a v(a)$. The rule computes

$$\begin{aligned} h_g &= \max\{\text{range}(A_g), \text{median}_a\, u_{A,g}(a), 10^{-6}\}, \\ D_g &= \text{range}(P_g - A_g)/h_g, \\ M_g &= \{P_g(a_A) - P_g(a_P)\}/h_g, \\ U_g &= \{u_{P,g}(a_A) + u_{P,g}(a_P)\}/h_g, \\ \alpha_g &= \min\{1, \kappa/\max(D_g, 10^{-12})\}\mathbf{1}\{M_g \geq \zeta U_g\}. \end{aligned} \quad (12)$$

The scale h_g normalizes model disagreement D_g, the pairwise margin M_g, and the combined error score U_g; $\mathbf{1}\{\cdot\}$ equals one when its condition holds and zero otherwise. The cap $\kappa \in \{0.25, 0.5, 1, 2, 4\}$ and error multiplier $\zeta \in \{0, 0.25, 0.5, 1\}$ are selected on target validation data. If the indicator fails, the pooled-source predictor uses A_g. Otherwise,

$$\text{range}(S_g - A_g) = \alpha_g \,\text{range}(P_g - A_g) \leq \kappa h_g. \quad (13)$$

Score-cap derivation. To verify (13), note that $S_g - A_g = \alpha_g(P_g - A_g)$ and $\alpha_g \geq 0$,

$$\text{range}(S_g - A_g) = \alpha_g h_g D_g \leq \kappa h_g,$$

where the last step follows from (12), including $D_g = 0$.

I. Original pooled-source blend selection

In the original blend-selection procedure, two target simulation seeds supply adaptation groups. In each validation fold, both predictors are fitted on all source examples and one target seed, then evaluated on the other target seed. Regret is first averaged over groups within each demand scenario, then equally over scenarios and the two folds. A candidate blend must not worsen mean validation regret relative to A_g, and its regret increase in either complete-seed fold must be at most 0.1. Among admissible candidates, we select the one with lowest mean regret, using deterministic tie breaking. The pure reference predictor remains available. Los Angeles selects $(\kappa, \zeta) = (1, 0.25)$, while Jinan selects constant $\alpha_g = 0.9$.

After selection, both predictors are fitted again on the source data and all target adaptation groups. The fold-specific models are discarded. Online control extracts traffic features, scores every feasible phase with (9), and passes the minimizing phase to the common executor. There is no online retraining, simulator branching, or uncertainty-based guard in this original controller. Sparse coordination still returns the pressure reference for blocked proposals. The pressure reference is still computed to define the score origin.

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